

# A PROOF OF THE MOST INFORMATIVE BOOLEAN FUNCTION CONJECTURE

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**ABSTRACT.** We give a self-contained account of the reduction of the general Courtade–Kumar Boolean-function conjecture to a single low-dimensional inequality. We follow the differential-equation approach introduced earlier by some of the authors and then prove three lower bounds on a moment functional, substantially reducing the dimensionality of the original formulation. The proof of the final low-dimensional inequality, which uses a careful partition of the region and a combination of analytic and interval-arithmetic bounds, constitutes the bulk of the full paper.

## 1. INTRODUCTION

For prior work and an introduction to the problem, please see the [full version](#).

**1.1. The conjecture.** Let  $X$  be uniform on the Boolean cube  $\{-1, 1\}^n$ , and let  $Y$  be obtained from  $X$  by passing each coordinate independently through a binary symmetric channel with crossover probability  $p \in (0, \frac{1}{2})$ ; equivalently,  $Y = X \circ Z$ , where the  $Z_i$  are independent signs, independent of  $X$ , with  $\mathbb{P}(Z_i = -1) = p$ , and  $\circ$  denotes coordinatewise multiplication. For a Boolean function  $g : \{-1, 1\}^n \rightarrow \{0, 1\}$ , one asks how much information a single output bit  $g(X)$  can carry about the noisy observation  $Y$ . Courtade and Kumar conjectured that the dictator functions  $g(x) = \frac{1+x_i}{2}$  are optimal:

$$I(g(X); Y) \leq \ln 2 - h(p) \quad \text{for every Boolean } g, \tag{1}$$

where  $h(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy in nats. Define

$$H_2(t) = \frac{h(t)}{\ln 2}, \quad I_2(A; C) = \frac{I(A; C)}{\ln 2}.$$

Dividing (1) by  $\ln 2$  gives the familiar inequality in bits:

$$I_2(g(X); Y) \leq 1 - H_2(p).$$

The bound is attained by any dictator, for which  $g(X)$  is a single input coordinate encoded as a bit.

Writing  $(T_\rho g)(y) = \mathbb{E}[g(X) \mid Y = y]$ , with  $\rho = 1 - 2p$ , the mutual information can be expressed through the noise operator as

$$I(g(X); Y) = h(\mathbb{P}(g(X) = 1)) - \mathbb{E}\left[h((T_\rho g)(Y))\right],$$

so (1) is a sharp entropy inequality for the Boolean noise operator. The nonlinear entropy functional and the possibility of contributions from all Fourier levels of  $g$  distinguish this problem from more classical quadratic extremal questions, such as noise stability.

**1.2. The differential-equation approach and this paper.** The present work builds on the differential-equation approach of Chen, Gohari, and Nair [12], itself a limiting form of the auxiliary-receiver method of Gohari and Nair [8] using a continuum of degraded receivers. One evolves the channel along a path of binary symmetric channels with time-varying crossover probability and differentiates the mutual information  $I(g(X); Y_t)$  along the path. Combined with induction on the dimension, this reduces the conjecture to a *pointwise* functional inequality—a Bellman inequality—for a candidate function of local moments. For the balanced case, [12] reduced the problem to an inequality in fourteen variables and proposed four inequalities, each involving at most four variables, that together would imply it. One of those four-variable inequalities, however, turns out to have violations near the boundary of the feasible region.

This paper isolates the required inequality as cleanly as possible. Section 2 fixes the notation, defines the hybrid candidate  $B$ , and introduces the unrestricted edge cost  $\zeta$  and the Bellman difference  $R_B$ . Section 3 immediately states the reduced hybrid inequality  $\zeta \geq R_B$  and proves that it implies the full Courtade–Kumar conjecture. Sections 4, 5, and 6 then define and prove the three key lower bounds on  $\zeta$ , reducing the outstanding step to a low-dimensional inequality, which is then validated with substantial assistance from AI. For each lower bound, we also describe the respective roles of humans and AI in its development.

**The main new ingredients.** Beyond the differential-equation framework and its balanced candidate  $\phi$ , the present work contributes: (i) two new explicit lower bounds on the unrestricted edge value  $\zeta$ , the log-sum bound  $L_2$  and the tangent bound  $L_3$ , which repair the boundary failure of the previously proposed bound and were developed through extensive interaction with artificial-intelligence systems, used for symbolic and numerical exploration, with every statement and proof checked and completed by the authors; and (ii) the identification of the hybrid candidate  $B = \max\{\phi, \psi\}$ , which closes the general unbalanced case and was conceived entirely by the human authors.

## 2. NOTATION AND THE CANDIDATE FUNCTIONS

Throughout the paper, entropies, divergences, edge costs, and mutual information quantities are measured in *nats*;  $\ln$  denotes the natural logarithm. This single convention is used everywhere, so that no normalisation constant intervenes between the edge inequalities of Sections 4, 5, and 6 and the candidate  $B$  defined below.

**2.1. Entropy, its derivative, and its inverse.** Define the binary entropy function and its derivative by

$$h(t) = -t \ln t - (1-t) \ln(1-t), \quad J(t) = h'(t) = \ln \frac{1-t}{t}, \quad 0 < t < 1, \quad (2)$$

with the continuous values  $h(0) = h(1) = 0$ . The inverse

$$\iota : [0, \ln 2] \rightarrow [0, \tfrac{1}{2}], \quad \iota(e) = h^{-1}(e), \quad (3)$$

always denotes the *lower* branch of the inverse binary entropy function.

**2.2. The edge cost.** The binary relative entropy in nats is

$$D(x\|y) = x \ln \frac{x}{y} + (1-x) \ln \frac{1-x}{1-y}, \quad x, y \in (0, 1), \quad (4)$$

with its standard lower-semicontinuous extension to  $[0, 1]^2$ . The *symmetrised edge cost* is

$$j(u, w) = \tfrac{1}{2}(u - w)(J(w) - J(u)) = \tfrac{1}{2}(D(u\|w) + D(w\|u)) \geq 0. \quad (5)$$

We use the lower-semicontinuous extension of  $j$  to  $[0, 1]^2$ , for which

$$j(0, 0) = j(1, 1) = 0, \quad j(u, w) = +\infty \quad \text{at every other boundary point.} \quad (6)$$

**2.3. The endpoint profile  $\eta$  and the perspective  $F$ .** For  $0 < e \leq \ln 2$ , define the *endpoint profile*

$$\eta(e) = (1 - 2\iota(e))J(\iota(e)), \quad \eta(\ln 2) = 0. \quad (7)$$

Define  $\mathcal{L}(t) = \frac{2h(t)}{1-2t}$  for  $0 \leq t < \frac{1}{2}$ ; it is a strictly increasing bijection from  $[0, \frac{1}{2})$  onto  $[0, \infty)$ . For  $s > 0$  and  $e > 0$ , let  $v = v(s, e) \in (0, \frac{1}{2})$  be the unique solution of  $\mathcal{L}(v) = 2e/s$ , equivalently,

$$sh(v) = e(1 - 2v), \quad (8)$$

and define the *perspective of the cost*

$$F(s, e) = sJ(v(s, e)), \quad F(0, e) = 0. \quad (9)$$

We extend  $F$  evenly in its first argument,  $F(-s, e) = F(s, e)$ , and set the lower-semicontinuous zero-entropy values  $F(0, 0) = 0$  and  $F(s, 0) = +\infty$  for  $s \neq 0$ . Equivalently, define

$$\Gamma(z) = zJ(\mathcal{L}^{-1}(1/z)) \quad (z > 0), \quad \Gamma(0) = 0.$$

Then, for  $s \in \mathbb{R}$  and  $e > 0$ ,

$$F(s, e) = 2e\Gamma\left(\frac{|s|}{2e}\right).$$

**2.4. The candidates  $\Phi, \phi, \psi, B$ .** For  $s \geq 0$  and  $0 < e \leq \ln 2$ , put

$$\Phi(s, e) = \eta(e) - F(s, e). \quad (10)$$

The *balanced candidate*, the *unbalanced candidate*, and their pointwise maximum, the *hybrid candidate*, are

$$\phi(m, e) = \Phi(|1 - 2m|, e) = \eta(e) - F(|1 - 2m|, e), \quad (11)$$

$$\psi(m, e) = \eta(e + \ln 2 - h(m)), \quad (12)$$

$$B(m, e) = \max\{\phi(m, e), \psi(m, e)\}, \quad (13)$$

evaluated on  $0 < m < 1$ ,  $0 < e \leq h(m)$ . The argument  $e + \ln 2 - h(m)$  of  $\eta$  in (12) lies in  $(0, \ln 2]$  because  $0 < e \leq h(m)$ . At the deterministic entropy cap, the contact identity (8) gives

$$\phi(m, h(m)) = 0, \quad \psi(m, h(m)) = \eta(\ln 2) = 0, \quad B(m, h(m)) = 0. \quad (14)$$

The role of  $\phi$  is to carry the sharp balanced estimate; the role of  $\psi$  is to absorb the entropy deficit  $\ln 2 - h(m)$  of an unbalanced output. We record the basic properties used later:  $\phi(1 - m, e) = \phi(m, e)$ ,  $\phi(\frac{1}{2}, e) = \eta(e)$ ,  $\phi \geq 0$  with equality if and only if  $e = h(m)$ , and  $e \mapsto \phi(m, e)$  is strictly convex on  $(0, h(m))$ ;  $\eta$  is strictly decreasing and strictly convex on  $(0, \ln 2)$ .

**2.5. Four-moment tuples, the edge value, and Bellman differences.** A *four-moment tuple* is  $M = (\mu_u, \mu_w, e_u, e_w)$  with  $0 \leq \mu_u, \mu_w \leq 1$ ,  $0 \leq e_u \leq h(\mu_u)$ , and  $0 \leq e_w \leq h(\mu_w)$ . We write  $\mathcal{M}^+$  for the subset with  $0 < \mu_u, \mu_w < 1$  and strictly positive entropy coordinates.

**Definition 2.1** (Edge value). Let  $\mathcal{S}(M)$  be the set of joint laws of  $(U, W)$  on  $[0, 1]^2$  with

$$\mathbb{E}U = \mu_u, \quad \mathbb{E}W = \mu_w, \quad \mathbb{E}h(U) = e_u, \quad \mathbb{E}h(W) = e_w. \quad (15)$$

The *unrestricted edge value* of  $M$  is

$$\zeta(M) = \inf_{(U, W) \in \mathcal{S}(M)} \mathbb{E}[j(U, W)] \in [0, +\infty], \quad (16)$$

with the boundary conventions (6). No pointwise ordering of  $U$  and  $W$  is imposed.

**Lemma 2.2** (Feasibility).  $\mathcal{S}(M) \neq \emptyset$  if and only if  $0 \leq \mu_u, \mu_w \leq 1$ ,  $0 \leq e_u \leq h(\mu_u)$ , and  $0 \leq e_w \leq h(\mu_w)$ .

*Proof.* Necessity follows from the nonnegativity and concavity of  $h$ . Conversely, fix  $0 < \mu < 1$  and  $0 \leq e \leq h(\mu)$ , and set  $\lambda = e/h(\mu)$ . Let  $V = \mu$  with probability  $\lambda$  and, with probability  $1 - \lambda$ , let  $V$  be Bernoulli with mean  $\mu$ . Then  $\mathbb{E}V = \mu$  and  $\mathbb{E}h(V) = \lambda h(\mu) = e$ . If  $\mu = 0$  or  $\mu = 1$ , feasibility forces  $e = 0$ , and we take  $V = \mu$  almost surely. Construct the two marginals by these rules and couple them independently.  $\square$

The relabelling  $(U, W) \mapsto (W, U)$  maps  $\mathcal{S}(\mu_u, \mu_w, e_u, e_w)$  bijectively onto  $\mathcal{S}(\mu_w, \mu_u, e_w, e_u)$  and preserves the edge cost. Consequently,

$$\zeta(\mu_u, \mu_w, e_u, e_w) = \zeta(\mu_w, \mu_u, e_w, e_u), \quad (17)$$

and in any lower bound for  $\zeta$ , we may assume  $\mu_u \leq \mu_w$ . For any candidate  $A \in \{\phi, \psi, B\}$ , its *Bellman difference* at  $M = (\mu_u, \mu_w, e_u, e_w)$  is

$$R_A(M) = A\left(\frac{\mu_u + \mu_w}{2}, \frac{e_u + e_w}{2}\right) - \frac{1}{2}A(\mu_u, e_u) - \frac{1}{2}A(\mu_w, e_w). \quad (18)$$

### 3. REDUCTION TO THE HYBRID BELLMAN INEQUALITY

**3.1. Historical motivation.** The differential-equation framework of [12] introduced the balanced candidate  $\phi$  defined in (11). That framework showed that the balanced Courtade–Kumar conjecture would follow from the pointwise Bellman inequality

$$\zeta(M) \geq R_\phi(M) \quad \text{for every } M \in \mathcal{M}^+. \quad (19)$$

The present paper does not use (19) as its global target. Instead, it works directly with the hybrid candidate  $B = \max\{\phi, \psi\}$  and proves the alternative, but sufficient, inequality

$$\zeta(M) \geq R_B(M). \quad (20)$$

The role of  $\phi$  is therefore local rather than global: it is used on the part of the feasible region where the parent value of  $B$  is attained by the  $\phi$ -branch. An analogous role is played by  $\psi$  on the complementary parent branch.

The next subsection states the hybrid inequality as the central reduced theorem and immediately explains why it implies the full Courtade–Kumar conjecture. This makes the destination of the subsequent lower-bound arguments explicit before any of their technical details are introduced.

### 3.2. The reduced theorem.

**Theorem 3.1** (Global hybrid Bellman inequality). *For every positive-entropy feasible tuple*

$$M = (\mu_u, \mu_w, e_u, e_w), \quad 0 < \mu_u, \mu_w < 1, \quad 0 < e_u \leq h(\mu_u), \quad 0 < e_w \leq h(\mu_w),$$

*one has*

$$\zeta(M) \geq R_B(M). \quad (21)$$

Theorem 3.1 is the central reduced statement of the paper. It is dimension-free and involves only the four moments collected in  $M = (\mu_u, \mu_w, e_u, e_w)$ . We first show that it implies the full Courtade–Kumar conjecture. Sections 4, 5 and 6 then reduce its proof to a single explicit four-variable inequality.

**3.3. Unrestricted static induction and edge energy.** For every array  $v : \{-1, 1\}^n \rightarrow (0, 1)$ , define the edge energy

$$E_n(v) = \sum_{i=1}^n 2^{-(n-1)} \sum_{y_{-i}} j(v_i^-(y_{-i}), v_i^+(y_{-i})), \quad (22)$$

where  $y_{-i}$  ranges over the  $(n-1)$ -coordinate subcube, and  $v_i^-(y_{-i})$  and  $v_i^+(y_{-i})$  are the two values of  $v$  on the edge in direction  $i$  with endpoints labelled by  $y_i = -1$  and  $y_i = +1$ , respectively.

The hybrid Bellman inequality implies a pointwise lower bound on this edge energy.

**Lemma 3.2** (Unrestricted static induction). *Assume Theorem 3.1, i.e., the hybrid Bellman inequality  $\zeta(M) \geq R_B(M)$  for every  $M \in \mathcal{M}^+$ . Then, for every  $n \geq 0$  and every array  $v : \{-1, 1\}^n \rightarrow (0, 1)$ ,*

$$E_n(v) \geq B(\mathbb{E}v, \mathbb{E}h(v)), \quad (23)$$

where expectations are taken with respect to the uniform measure on  $\{-1, 1\}^n$ , and  $E_0(v) = 0$ .

*Proof.* Both candidates vanish at the entropy cap:

$$\begin{aligned} \phi(m, h(m)) &= \Phi(|1 - 2m|, h(m)) = 0, \\ \psi(m, h(m)) &= \eta(\ln 2) = 0, \quad B(m, h(m)) = 0. \end{aligned}$$

For the first identity, if  $m < \frac{1}{2}$ , the defining contact point for  $F$  is  $v = m$ ; the other half follows by complementation, with the continuous midpoint value. Thus, the assertion is true for  $n = 0$ .

Assume the claim for dimension  $n - 1$  and split the last coordinate. Let  $a, b$  be the two slice means and  $e, f$  their mean entropies. The energy  $E_n(v)$  is the average of the two slice energies plus the expected cross-edge cost. By induction,

$$E_{n-1}(v|_{x_n=-1}) \geq B(a, e), \quad E_{n-1}(v|_{x_n=+1}) \geq B(b, f).$$

Averaging gives  $[B(a, e) + B(b, f)]/2$  as the contribution from the slices.

The unrestricted law of the pair of slice values yields a cross-edge cost of at least  $\zeta(M)$ , where  $M$  is the four-moment tuple associated with the two slices. The hybrid Bellman inequality gives  $\zeta(M) \geq R_B(M)$ , so the cross term contributes at least  $R_B(M)$ . Cancellation in the Bellman difference,

$$R_B(M) = B\left(\frac{a+b}{2}, \frac{e+f}{2}\right) - \frac{1}{2}(B(a, e) + B(b, f)),$$

shows that the total energy  $E_n(v)$  is at least  $B(\frac{a+b}{2}, \frac{e+f}{2})$ , i.e.,  $B(\mathbb{E}v, \mathbb{E}h(v))$ . Every slice has an interior mean and positive entropy because the original array takes values in  $(0, 1)$ , so no boundary case arises. Ordering the two slices is unnecessary.  $\square$

**3.4. Regularisation and the differential inequality along the flow.** We now apply Lemma 3.2 to the posterior profile along the noise flow. Fix  $0 < \varepsilon < \frac{1}{2}$  and regularise by setting

$$h_\varepsilon(x) = \varepsilon + (1 - 2\varepsilon)g(x), \quad m_\varepsilon = \mathbb{E}h_\varepsilon, \quad v_{\varepsilon,t} = T_{e^{-2t}}h_\varepsilon, \quad \gamma_\varepsilon(t) = \mathbb{E}h(v_{\varepsilon,t}). \quad (24)$$

Here,  $T_\rho$  is the noise operator with coordinate crossover probability  $(1 - \rho)/2$ . Every entry of  $v_{\varepsilon,t}$  lies in  $[\varepsilon, 1 - \varepsilon]$ . The generator at correlation  $e^{-2t}$  is

$$\sum_i (v_{\varepsilon,t}(x^{(i)}) - v_{\varepsilon,t}(x)),$$

where  $x^{(i)}$  is the neighbour obtained by flipping the  $i$ th coordinate. Pairing the two orientations of each cube edge gives the exact identity

$$\gamma'_\varepsilon(t) = E_n(v_{\varepsilon,t}), \quad (25)$$

with  $E_n$  as in (22). The factor  $2^{-(n-1)}$  and the factor  $\frac{1}{2}$  in the definition of  $j$  agree with this generator normalisation.

Applying Lemma 3.2 to  $v_{\varepsilon,t}$  and using the definitions of  $m_\varepsilon$  and  $\gamma_\varepsilon(t)$  shows that

$$\gamma'_\varepsilon(t) \geq B(m_\varepsilon, \gamma_\varepsilon(t)) \geq \psi(m_\varepsilon, \gamma_\varepsilon(t)) = \eta(\gamma_\varepsilon(t) + \ln 2 - h(m_\varepsilon)), \quad (26)$$

where the last inequality uses the pointwise bound  $B(m, e) \geq \psi(m, e)$ , and the equality uses the definition

$$\psi(m, e) = \eta(e + \ln 2 - h(m))$$

adapted to the regularised flow. Thus, replacing  $B$  by  $\psi$  yields a *scalar* differential inequality driven by  $\eta$ .

Write

$$\delta_\varepsilon(t) = \gamma_\varepsilon(t) + \ln 2 - h(m_\varepsilon). \quad (27)$$

The mean  $m_\varepsilon$  is preserved along the flow, so  $\delta'_\varepsilon(t) = \gamma'_\varepsilon(t)$ , and (26) becomes

$$\delta'_\varepsilon(t) \geq \eta(\delta_\varepsilon(t)). \quad (28)$$

Since  $m_\varepsilon \in [\varepsilon, 1 - \varepsilon]$ , the binary entropy satisfies

$$h(\varepsilon) \leq \delta_\varepsilon(0) = h(\varepsilon) + \ln 2 - h(m_\varepsilon) \leq \ln 2,$$

and the concavity of  $h$  gives  $\delta_\varepsilon(t) \leq \ln 2$  at every time. If  $\delta_\varepsilon(t)$  reaches  $\ln 2$ , then  $\gamma_\varepsilon(t) = h(m_\varepsilon)$ , and the mutual information is zero.

For  $0 < z < \ln 2$ , define

$$A_\varepsilon(z) = \int_{h(\varepsilon)}^z \frac{dr}{\eta(r)} = \frac{1}{2} \ln \frac{1 - 2\varepsilon}{1 - 2\iota(z)}. \quad (29)$$

Before  $\delta_\varepsilon$  reaches  $\ln 2$ ,

$$\frac{d}{dt} A_\varepsilon(\delta_\varepsilon(t)) \geq 1, \quad A_\varepsilon(\delta_\varepsilon(0)) \geq 0.$$

Therefore,

$$\delta_\varepsilon(t) \geq h\left(\frac{1 - e^{-2t}(1 - 2\varepsilon)}{2}\right).$$

Let  $G_\varepsilon = g(X) \text{ XOR } N_\varepsilon$ , where  $N_\varepsilon$  is an independent Bernoulli( $\varepsilon$ ) bit. Then

$$I(G_\varepsilon; Y_t) = h(m_\varepsilon) - \gamma_\varepsilon(t) = \ln 2 - \delta_\varepsilon(t) \leq \ln 2 - h\left(\frac{1 - e^{-2t}(1 - 2\varepsilon)}{2}\right).$$

Letting  $\varepsilon \downarrow 0$  and using the continuity of finite-alphabet entropy, we obtain the bound for the original function  $g$  and the noise level  $p = \frac{1 - e^{-2t}}{2}$ :

$$I(g(X); Y) \leq \ln 2 - h(p) \quad \text{for } 0 < p < \frac{1}{2}.$$

At  $p = 0$ , we have  $I(g(X); X) = h(\mathbb{P}(g(X) = 1)) \leq \ln 2$ . At  $p = \frac{1}{2}$ , the output  $Y$  is independent of  $X$ , so the mutual information vanishes. Complementing every output coordinate reduces  $p > \frac{1}{2}$  to  $1 - p$  without changing the mutual information or  $h(p)$ .

We have therefore proved the following.

**Theorem 3.3** (General Courtade–Kumar conjecture). *Assume the hybrid Bellman inequality (21). Then, for every Boolean function  $g$  and every  $p \in (0, \frac{1}{2})$ ,*

$$I(g(X); Y) \leq \ln 2 - h(p),$$

where  $Y$  is obtained from  $X$  by passing each coordinate independently through a binary symmetric channel with crossover probability  $p$ . Equivalently, in bits,

$$I_2(g(X); Y) \leq 1 - H_2(p).$$

**3.5. Plan of the proof of Theorem 3.1.** It remains to prove the hybrid Bellman inequality (21). The proof uses three explicit lower bounds on the edge value, namely the log-sum bound  $L_2(M)$ , the tangent bound  $L_3(M)$ , and the perspective bound  $L_1(M)$ , and establishes

$$\zeta(M) \geq L_2(M), \quad \zeta(M) \geq L_3(M), \quad \zeta(M) \geq L_1(M)$$

on their stated domains. The log-sum bound  $L_2$  and the tangent bound  $L_3$  are the two new boundary estimates. The perspective bound  $L_1$  is built from the residual function  $\kappa$  inherited from the balanced programme. A same-side estimate  $L_{\text{corn}}$  is also established along the way and is used as an auxiliary ingredient in the convex transfer, but it does not enter the final maximum.

After these bounds have been established, we set

$$L_{\max}(M) = \max\{L_1(M), L_2(M), L_3(M)\},$$

with the convention that  $L_3(M) = -\infty$  outside the domain on which its contact triple exists. Theorem 3.1 then follows from the four-variable comparison

$$L_{\max}(M) \geq R_B(M).$$

Sections 4, 5, and 6 define and prove the lower bounds.

We next define and prove the lower bounds announced in Section 3.5. We first introduce the residual  $\kappa$  and the perspective bound  $L_1$ . The reflection and entropy-coordinate convexity properties of  $\kappa$  are then proved, making  $L_1$  an unconditional lower bound. We next establish the log-sum bound  $L_2$  and, finally, the tangent bound  $L_3$ . Together, these estimates control the interior and boundary regimes of the feasible region.

#### 4. THE FIRST LOWER BOUND

**Remark 4.1.** This lower bound on  $\zeta$  was conceived in [12] and was stated there as a conditional lower bound, subject to the validity of Conjecture 4. Proposition 4.2 proves that conjecture, thus establishing the lower bound unconditionally. The proof of Proposition 4.2, which entails proving three two-dimensional inequalities, was developed predominantly by AI, with minimal but non-zero human guidance. This lower bound is Theorem 3.1 in the full paper.

**4.1. The residual  $\kappa$  and the perspective bound  $L_1$ .** The function  $F$  defined in (9) is the perspective of the edge cost in the mean-difference variable at fixed average entropy. Define the residual function

$$\kappa(u, w) = j(u, w) - F\left(|u - w|, \frac{h(u) + h(w)}{2}\right) \quad (30)$$

and the corresponding entropy-coordinate function

$$g(e_1, e_2) = \kappa(\iota(e_1), \iota(e_2)). \quad (31)$$

Its lower-semicontinuous closure satisfies

$$g(0, 0) = 0, \quad g(e, e) = 0, \quad g(0, e) = g(e, 0) = +\infty \quad (e > 0).$$

For a feasible four-moment tuple  $M = (\mu_u, \mu_w, e_u, e_w)$ , define

$$L_1(M) = F\left(|\mu_w - \mu_u|, \frac{e_u + e_w}{2}\right) + g(e_u, e_w). \quad (32)$$

The proof that  $L_1$  is a lower bound on  $\zeta$  rests on the following two properties of  $\kappa$ .

**Proposition 4.2** (Reflection and entropy-coordinate convexity). *The residual  $\kappa$  has the following properties.*

(i) *Reflection property: for  $u, w \in (0, \frac{1}{2}]$ ,*

$$\kappa(u, w) \leq \kappa(1 - u, w).$$

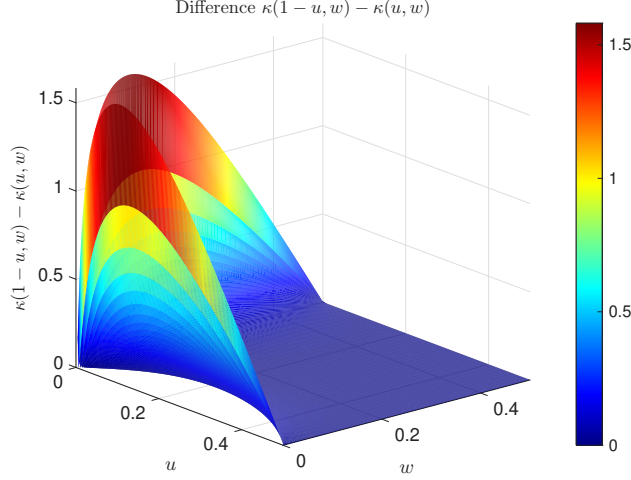
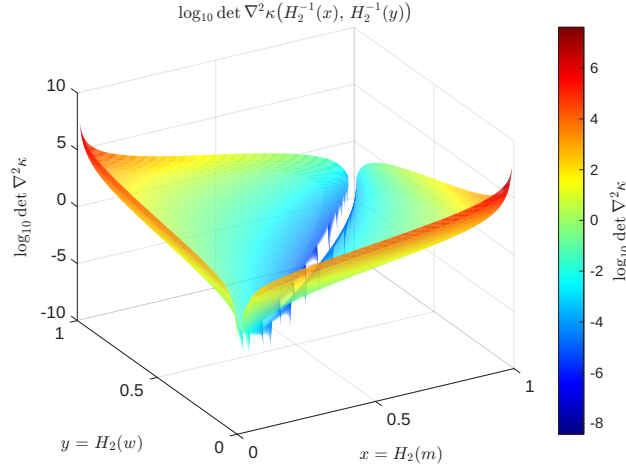
(ii) *Joint convexity property: the function*

$$(e_1, e_2) \mapsto \kappa(\iota(e_1), \iota(e_2))$$

*is jointly convex on  $(0, \ln 2)^2$ , and its lower-semicontinuous extended-real closure is convex on  $[0, \ln 2]^2$ .*

*Proof.* The proof details of this, a combination of analytic and interval arithmetic ideas, can be found in the full version. Observe that the two assertions of Proposition 4.2 involve only two variables. We just provide the 3-D plots of the functions that provide evidence to the assertions. For illustration, Figure 1 plots  $\kappa(1 - u, w) - \kappa(u, w)$  for  $u, w \in (0, 0.5]$ . Figure 2 plots the determinant of the Hessian of  $\kappa(\iota(u), \iota(w))$ , and Figure 3 plots  $\frac{\partial^2}{\partial u^2} \kappa(\iota(u), \iota(w))$ . □



FIGURE 1.  $\kappa(1-u, w) - \kappa(u, w)$ .FIGURE 2. The determinant of the Hessian of  $\kappa(\iota(u), \iota(w))$ .

The two assertions of Proposition 4.2 imply

$$\zeta(M) \geq L_1(M) \quad (33)$$

for every feasible tuple; this is stated formally as Theorem 4.3. It is important that (33) is only a lower bound on  $\zeta$ . The stronger comparison

$$L_1(M) \geq R_\phi(M)$$

proposed in the earlier balanced programme is false near the boundary of the feasible region. The two additional bounds  $L_2$  and  $L_3$ , introduced in Sections 5 and 6, are designed to control precisely those boundary regions. The eventual comparison is therefore made with  $R_B$ , using the maximum  $L_{\max} = \max\{L_1, L_2, L_3\}$ , rather than by attempting to prove a global inequality involving  $R_\phi$  alone.

**Theorem 4.3** ([12]). *For every feasible four-moment tuple, with the boundary conventions of (6),  $\zeta(M) \geq L_1(M)$ .*

*Proof.* Assume first that  $e_u, e_w > 0$ , and let  $(U, W)$  be any representing law. If  $\mathbb{E}j(U, W) = +\infty$ , the bound is trivial; otherwise, almost surely,  $(U, W) \in (0, 1)^2$  or  $(U, W) \in \{(0, 0), (1, 1)\}$ . For



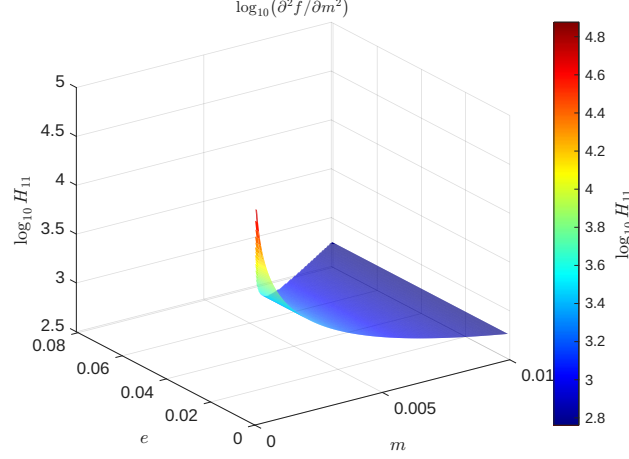


FIGURE 3. The surface of  $\frac{\partial^2}{\partial u^2} \kappa(\iota(u), \iota(w))$ .

interior points, the reflection property (Proposition 4.2(i)) and the definition of  $g$  give

$$j(U, W) \geq F\left(|U - W|, \frac{h(U) + h(W)}{2}\right) + g(h(U), h(W)),$$

using the even extension of  $F$ . The same holds at  $(0, 0)$  and  $(1, 1)$ , since all three terms vanish. The even extension of  $F$  is jointly convex, by convexity in the nonnegative radial argument together with monotonicity, and its zero-entropy closure is convex. The convexity of  $g$  is Proposition 4.2(ii). Applying Jensen's inequality to  $F$  and to the convex closure of  $g$ , we obtain

$$\mathbb{E}j(U, W) \geq F(|\mu_w - \mu_u|, \frac{e_u + e_w}{2}) + g(e_u, e_w) = L_1(M).$$

Taking the infimum over representing laws proves the positive-entropy case.

*Zero-entropy boundary.* If  $e_u = e_w = 0$ , then  $U, W \in \{0, 1\}$  almost surely. Finite cost forces  $U = W$  almost surely, so  $\zeta(M) = L_1(M) = +\infty$  if  $\mu_u \neq \mu_w$ , while the coupling  $U = W$  with the prescribed Bernoulli mean gives  $\zeta(M) = L_1(M) = 0$  if  $\mu_u = \mu_w$ . If exactly one entropy vanishes, the corresponding variable is endpoint-valued almost surely, while the other is interior with positive probability, forcing infinite edge cost. The closure of  $g$  gives  $L_1(M) = +\infty$  as well.  $\square$

### Refutation of Conjecture 5 in [12] and its implications.

Conjecture 5 in [12] postulated that  $L_1(M) \geq R_\phi(M)$ . If true, it would have proved the Courtade–Kumar conjecture for balanced functions. However, counterexamples were discovered using AI near the boundary of the feasible region. One set of counterexamples occurred when both  $\mu_u$  and  $\mu_w$  were close to zero or, symmetrically, both were close to one. Another set occurred when  $\mu_u$  was close to zero and  $\mu_w$  was close to one or, symmetrically,  $\mu_u$  was close to one and  $\mu_w$  was close to zero. However, different human-derived lower bounds, one for each regime, were developed to show that  $\zeta(M) \geq R_\phi(M)$  even at those points. The two lower bounds developed by AI that follow serve a similar purpose.

## 5. THE SECOND LOWER BOUND

**Remark 5.1.** The function  $j(u, w) = \frac{1}{2}(D(u\|w) + D(w\|u))$  is jointly convex in  $(u, w)$ . Taking expectations and then the infimum over representing laws immediately gives

$$\zeta(M) \geq \frac{1}{2}(D(\mu_u\|\mu_w) + D(\mu_w\|\mu_u)).$$

The authors used this lower bound as an intermediate quantity between  $\zeta(M)$  and  $R_\phi(M)$  when both  $\mu_u$  and  $\mu_w$  are close to zero. The bound presented below, developed by AI, uses joint convexity

and provides a highly nontrivial lower bound on the Jensen gap. This lower bound unifies the arguments in Theorem 11.2, and the arguments in sections S.26.1 and S.26.13 of the main paper.

**5.1. Second lower bound: the log-sum bound  $L_2$ .** Let  $\nu_x = x(1-x)$  and  $V(x, y) = \max\{x, y\}(1 - \min\{x, y\})$ . Define

$$\Upsilon(z) = \min \left\{ \frac{z}{-(1-z)\ln(1-z)}, \frac{1-z}{-z\ln z} \right\}, \quad 0 < z < 1. \quad (34)$$

It is immediate that  $\Upsilon(z) \geq 1$ , since each term is bounded below by one.

**Theorem 5.2** (Global log-sum lower bound). *Let  $(a, b) \in (0, 1)^2$ . For every interior feasible four-moment tuple  $M = (\mu_u, \mu_w, e_u, e_w)$ , we have*

$$\begin{aligned} \zeta(M) &\geq j(a, b) + j_x(a, b)(\mu_u - a) + j_y(a, b)(\mu_w - b) \\ &\quad + \frac{(a-b)^2}{4V(a, b)} \left[ \Upsilon(a)(-\mu_u \ln a - (1 - \mu_u) \ln(1 - a) - e_u) \right. \\ &\quad \left. + \Upsilon(b)(-\mu_w \ln b - (1 - \mu_w) \ln(1 - b) - e_w) \right]. \end{aligned}$$

*Proof.* Fix interior anchors  $a, b$ , and consider the difference between  $j$  and its supporting plane at  $(a, b)$ :

$$Q_{a,b}(u, w) := j(u, w) - j(a, b) - j_x(a, b)(u - a) - j_y(a, b)(w - b). \quad (35)$$

For positive masses that are not necessarily normalised, define

$$d_{\text{sc}}(x\|y) := x \ln(x/y) - x + y. \quad (36)$$

Observe that  $d_{\text{sc}}$  is the perspective of  $x \ln x - x + 1$  and hence is jointly convex in  $(x, y)$ .

Set

$$r_1 := \frac{u}{a}, \quad r_0 := \frac{1-u}{1-a}, \quad t_1 := \frac{w}{b}, \quad t_0 := \frac{1-w}{1-b}. \quad (37)$$

A simple manipulation yields the identity

$$\begin{aligned} 2Q_{a,b}(u, w) &= a d_{\text{sc}}(r_1\|t_1) + (1-a) d_{\text{sc}}(r_0\|t_0) \\ &\quad + b d_{\text{sc}}(t_1\|r_1) + (1-b) d_{\text{sc}}(t_0\|r_0). \end{aligned} \quad (38)$$

The log-sum inequality, equivalently the joint convexity of  $d_{\text{sc}}$ , applied separately to the two rows in (38), yields

$$Q_{a,b}(u, w) \geq \frac{1}{2} (d_{\text{sc}}(1\|T) + d_{\text{sc}}(1\|S)), \quad (39)$$

where

$$T := 1 + \frac{(a-b)(w-b)}{\nu_b}, \quad S := 1 - \frac{(a-b)(u-a)}{\nu_a}. \quad (40)$$

For  $z \in (-1, 0)$ , we have

$$z - \ln(1+z) \geq \frac{z^2}{2},$$

and for  $z > 0$ , we have

$$z - \ln(1+z) \geq \frac{z^2}{2(1+z)}.$$

Therefore, for  $z > -1$ ,

$$d_{\text{sc}}(1\|1+z) = z - \ln(1+z) \geq \frac{z^2}{2 \max\{1, 1+z\}}. \quad (41)$$

Since  $0 \leq u, w \leq 1$ ,

$$\max\{1, T\} \leq \frac{V(a, b)}{\nu_b}, \quad \max\{1, S\} \leq \frac{V(a, b)}{\nu_a}. \quad (42)$$

For example, if  $a \leq b$ , the relevant maxima are bounded by  $(1-a)/(1-b)$  and  $b/a$ , respectively. Combining (39)–(42) gives

$$Q_{a,b}(u, w) \geq \frac{(a-b)^2}{4V(a, b)} \left( \frac{(u-a)^2}{\nu_a} + \frac{(w-b)^2}{\nu_b} \right). \quad (43)$$

We now convert squared deviations to entropy deficits. Taylor's integral formula gives the continuous extension

$$\frac{D(x||a)}{(x-a)^2} = \int_0^1 \frac{1-t}{(a+t(x-a))(1-a-t(x-a))} dt. \quad (44)$$

For each fixed  $t \in [0, 1]$ , the integrand in (44) is convex as a function of  $x$ , because

$$\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$$

is convex on  $(0, 1)$ . Therefore, the ratio on the left-hand side of (44) is convex on  $[0, 1]$  and attains its maximum at an endpoint. Hence,

$$D(x||a) \leq \Gamma_a(x-a)^2, \quad \Gamma_a := \max \left\{ \frac{-\ln(1-a)}{a^2}, \frac{-\ln a}{(1-a)^2} \right\}. \quad (45)$$

The definitions give

$$\frac{1}{\nu_a \Gamma_a} = \Upsilon(a). \quad (46)$$

Consequently,

$$\frac{(x-a)^2}{\nu_a} \geq \Upsilon(a) D(x||a). \quad (47)$$

The analogous inequality holds with  $a$  replaced by  $b$ .

Therefore,

$$\begin{aligned} Q_{a,b}(u, w) &\geq \frac{(a-b)^2}{4V(a, b)} \left( \frac{(u-a)^2}{\nu_a} + \frac{(w-b)^2}{\nu_b} \right) \\ &\geq \frac{(a-b)^2}{4V(a, b)} (\Upsilon(a) D(u||a) + \Upsilon(b) D(w||b)). \end{aligned} \quad (48)$$

Thus, we have the pointwise inequality

$$\begin{aligned} j(u, w) &\geq j(a, b) + j_x(a, b)(u-a) + j_y(a, b)(w-b) \\ &\quad + \frac{(a-b)^2}{4V(a, b)} (\Upsilon(a) D(u||a) + \Upsilon(b) D(w||b)). \end{aligned}$$

Taking expectations over any law of  $(U, W)$  that satisfies the moment constraints  $M = (\mu_u, \mu_w, e_u, e_w)$  gives

$$\begin{aligned} \mathbb{E}j(U, W) &\geq j(a, b) + j_x(a, b)(\mu_u - a) + j_y(a, b)(\mu_w - b) \\ &\quad + \frac{(a-b)^2}{4V(a, b)} \left[ \Upsilon(a)(-\mu_u \ln a - (1-\mu_u) \ln(1-a) - e_u) \right. \\ &\quad \left. + \Upsilon(b)(-\mu_w \ln b - (1-\mu_w) \ln(1-b) - e_w) \right]. \end{aligned}$$

Taking the infimum over all such laws proves the claim.  $\square$

Fix  $M = (\mu_u, \mu_w, e_u, e_w) \in \mathcal{M}^+$ . Define the entropy deficits

$$I_{\mu_u} = h(\mu_u) - e_u, \quad I_{\mu_w} = h(\mu_w) - e_w, \quad s = \frac{1}{2}(I_{\mu_u} + I_{\mu_w}).$$

A useful special case is  $a = \mu_u$  and  $b = \mu_w$ . The following corollary is then immediate.

**Corollary 5.3** (Mean-centred log-sum lower bound). *For every  $M \in \mathcal{M}^+$ ,*

$$\zeta(M) \geq L_2(M) := j(\mu_u, \mu_w) + \frac{(\mu_u - \mu_w)^2}{4V(\mu_u, \mu_w)} (\Upsilon(\mu_u)I_{\mu_u} + \Upsilon(\mu_w)I_{\mu_w}). \quad (49)$$

*If the labels are arranged so that  $\mu_u \leq \mu_w$ , then, with  $d = \mu_w - \mu_u$  and*

$$\lambda = \frac{\min\{\Upsilon(\mu_u), \Upsilon(\mu_w)\}}{2\mu_w(1 - \mu_u)},$$

*one has  $\zeta(M) \geq j(\mu_u, \mu_w) + \lambda d^2 s$ . In particular, the weaker choice  $\lambda = (2\mu_w(1 - \mu_u))^{-1}$  is valid.*

**Remark 5.4.** While the mean-centred log-sum lower bound suffices for the balanced case, the more general bound is used in the calculations for 33,572 leaves in the boundary region and 247 leaves in the moderate-mean region.

## 6. THE THIRD LOWER BOUND $L_3$

**Remark 6.1.** In [12, Theorem 4], the authors explicitly computed  $\zeta(\mu_u, 1 - \mu_u, e_u, e_u)$ . The heart of the argument was that the only points needed to compute  $\zeta$  were  $(0,0)$ ,  $(1,1)$ , and a single point of the form  $(u, 1 - u)$ . The third lower bound, obtained by AI, can be seen as a significant generalisation of that result. This lower bound is present in Proposition 3.4, Theorem 3.10 of the main paper.

We first state a general lower-bound principle for a fixed anchor  $(a, b)$ , where  $0 < a < b < 1$ . In Section 6.1, we use it with two different choices of anchors to obtain two explicit lower bounds. The key idea is as follows: suppose that

$$j(x, y) \geq k + u_0 x + v_0 y - \alpha h(x) - \delta h(y) \quad (x, y \in [0, 1])$$

for some  $k, u_0, v_0, \alpha, \delta$ . This is a Lagrangian formulation associated with the moment constraints. Then every feasible tuple satisfies

$$\zeta(\mu_u, \mu_w, e_u, e_w) \geq k + u_0 \mu_u + v_0 \mu_w - \alpha e_u - \delta e_w.$$

This follows by integrating the pointwise inequality. The four moment constraints give the displayed affine expression. Taking the infimum over representing laws proves the assertion.

**Theorem 6.2** (Global contact-plane inequality). *Let  $0 < a < b < 1$  and set  $d = b - a$ . Let  $A, D$  solve*

$$\begin{pmatrix} -\ln a & -\ln b \\ -\ln(1-a) & -\ln(1-b) \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} d^2/(2ab) \\ d^2/[2(1-a)(1-b)] \end{pmatrix}. \quad (50)$$

*Assume that the solution satisfies*

$$A \geq 0, \quad D \geq 0.$$

*Define*

$$c = \frac{j(a, b) + Ah(a) + Dh(b)}{b - a}.$$

*Then  $c > 0$ , and*

$$Q_c(x, y) := j(x, y) - c(y - x) + Ah(x) + Dh(y) \geq 0 \quad (x, y \in [0, 1]). \quad (51)$$

*Equality holds at  $(0,0)$ ,  $(a,b)$ , and  $(1,1)$ . In fact, these are the only zeros of  $Q_c$ .*

*Proof.* Since  $A, D \geq 0$  and the right-hand side of (50) is nonzero, we have  $A + D > 0$ . Moreover,  $j(a, b) > 0$  because  $a \neq b$ . Hence,

$$c = \frac{j(a, b) + Ah(a) + Dh(b)}{b - a} > 0.$$

For arbitrary interior  $x, y$ , define

$$I_1(x, y) = A \ln x + D \ln y + \frac{(y - x)^2}{2xy},$$

$$I_0(x, y) = A \ln(1 - x) + D \ln(1 - y) + \frac{(y - x)^2}{2(1 - x)(1 - y)}.$$

Direct differentiation gives, for every real  $c$ ,

$$Q_c - x(Q_c)_x - y(Q_c)_y = -I_0, \quad (52)$$

$$Q_c + (1 - x)(Q_c)_x + (1 - y)(Q_c)_y = -I_1. \quad (53)$$

These follow, for example, by using

$$h(x) - xJ(x) = -\ln(1 - x), \quad h(x) + (1 - x)J(x) = -\ln x$$

in the differentiated formula for  $Q_c$ .

The coefficient system gives

$$I_0(a, b) = I_1(a, b) = 0.$$

The definition of  $c$  also gives  $Q_c(a, b) = 0$ . Equations (52)–(53) then imply

$$(Q_c)_x(a, b) = (Q_c)_y(a, b) = 0,$$

because the corresponding two linear equations have determinant  $a - b \neq 0$ .

Consider the quotient

$$\mathcal{Q}(x, y) = \frac{j(x, y) + Ah(x) + Dh(y)}{y - x}, \quad 0 < x < y < 1.$$

We next show that it tends to  $+\infty$  along every approach to the boundary of this open triangle.

Along the interior diagonal, its numerator tends to  $(A + D)h(x) > 0$ , while its denominator tends to zero.

Near  $(0, 0)$ , if  $x/y \rightarrow 0$ , then

$$\frac{j(x, y)}{y - x} = \frac{J(x) - J(y)}{2} \rightarrow +\infty.$$

If  $x/y$  stays bounded below, either  $A > 0$  or  $D > 0$  makes the entropy part divided by  $y - x$  diverge, since  $h(t) \sim t \ln(1/t)$ . The analogous argument near  $(1, 1)$  uses  $(1 - y)/(1 - x)$ .

Along all other approaches to the boundary, the displayed Jeffreys quotient diverges directly. Thus, the sublevel sets of  $\mathcal{Q}$  are compact inside the open triangle, and  $\mathcal{Q}$  attains its minimum there.

At any critical point of  $\mathcal{Q}$ , put  $c_* = \mathcal{Q}(x, y)$ . Differentiating the quotient gives

$$Q_{c_*} = 0, \quad \nabla Q_{c_*} = 0.$$

The identities above then give  $I_0 = I_1 = 0$ . Lemma 6.3 shows that there is at most one such critical point.

But  $(a, b)$  is already a critical point, with  $\mathcal{Q}(a, b) = c$ . Therefore, it is the unique global minimum, and

$$\mathcal{Q}(x, y) \geq c \quad (0 < x < y < 1).$$

Multiplying by  $y - x > 0$  proves (51) on this triangle.

On the other triangle, where  $y \leq x$ ,

$$Q_c(x, y) = j(x, y) + c(x - y) + Ah(x) + Dh(y) \geq 0.$$

In the open square, this expression cannot vanish when  $y \leq x$ , because  $A + D > 0$  and both entropies are positive.

At  $(0, 0)$  and  $(1, 1)$ , the expression is zero. At every other boundary point, its cost term is infinite.

Finally, any interior zero with  $x < y$  is a minimum of the smooth, nonnegative function  $Q_c$ . Its gradient vanishes, so the uniqueness lemma forces it to be  $(a, b)$ . This identifies the full zero set.  $\square$

**Lemma 6.3** (Uniqueness). *For  $a, b \geq 0$  with  $a + b > 0$ , define*

$$f_1(u, w) = a \ln u + b \ln w + \frac{(u - w)^2}{2uw}, \quad f_0(u, w) = a \ln(1 - u) + b \ln(1 - w) + \frac{(u - w)^2}{2(1 - u)(1 - w)}.$$

*Then the system  $f_1 = K_1$ ,  $f_0 = K_0$ , with  $K_0, K_1 \geq 0$ , has at most one solution satisfying  $0 < u < w < 1$ .*

*Proof.* Set  $x = u/w$  and  $y = (1 - w)/(1 - u)$ , so that  $x, y \in (0, 1)$ ,  $u = x(1 - y)/(1 - xy)$ , and  $w = (1 - y)/(1 - xy)$ . Let  $\Delta(z) = (1 - z)^2/(2z)$  and  $C = a + b$ . The two equations become

$$F_1(x, y) = \Delta(x) + a \ln x + C \ln \frac{1-y}{1-xy} = K_1, \quad F_0(x, y) = \Delta(y) + b \ln y + C \ln \frac{1-x}{1-xy} = K_0.$$

For fixed  $x$ , the equation  $F_1 = K_1$  determines at most one branch  $y = y(x)$ , through

$$w(x) = \exp\{(K_1 - \Delta(x) - a \ln x)/C\}, \quad u(x) = xw(x), \quad y(x) = \frac{1 - w(x)}{1 - u(x)}.$$

Writing  $L_{\text{red}}(x) = F_0(x, y(x)) - K_0$ , a direct calculation gives

$$L'_{\text{red}}(x) = \frac{(1 - y)(1 - xy)}{4Cx^2y^2(1 - x)} R(x, y),$$

with

$$R(x, y) = (1 - x^2)(1 - y^2) - 2(1 + xy)(ax + by) + 4abxy.$$

Thus, critical points of  $L_{\text{red}}$  along  $F_1 = K_1$  lie on the *fold*  $R = 0$ . We show that the fold does not meet  $\{F_1 \geq 0, F_0 \geq 0\}$ . Using  $\ln z < 2(z - 1)/(z + 1)$  and  $\ln z < z - 1$  on  $(0, 1)$ , the constraints  $F_1, F_0 \geq 0$  imply  $L_1, L_2 \geq 0$ , where

$$\begin{aligned} L_1 &= (1 - x^2)(1 - xy) - 2ax(2 + y - xy) - 2bxy(1 + x), \\ L_2 &= (1 - y^2)(1 - xy) - 2axy(1 + y) - 2by(2 + x - xy). \end{aligned}$$

Put  $\mathcal{D} = 1 + xy$ ,  $\Sigma = x + y$ ,  $P = \mathcal{D} - 2ax$ , and  $Q = \mathcal{D} - 2by$ , so that  $R = PQ - \Sigma^2$ . On the fold,  $PQ = \Sigma^2 > 0$ , and the identity

$$L_1 = (2 + y - xy)P + x(1 + x)Q - (1 + y)(2x^2 + xy + x + 1 - x^2y)$$

excludes  $P, Q < 0$ , so  $P, Q > 0$ . With  $P = t\Sigma$ ,  $Q = \Sigma/t$ , and  $\Sigma/\mathcal{D} \leq t \leq \mathcal{D}/\Sigma$ , as follows from  $P, Q \leq \mathcal{D}$ , the convexity of  $t + t^{-1}$  gives

$$\begin{aligned} (1 - y)L_1 + (1 - x)L_2 &= (1 - xy)[2\Sigma(t + t^{-1}) + M] \\ &\leq (1 - xy)\left[2\Sigma\left(\frac{\mathcal{D}}{\Sigma} + \frac{\Sigma}{\mathcal{D}}\right) + M\right] \\ &= -\frac{(1 - x)(1 - y)(1 - xy)^2(x + y)}{1 + xy} < 0, \end{aligned}$$

with  $M = x^2y - x^2 + xy^2 - 4xy - x - y^2 - y - 2$ , contradicting  $L_1, L_2 \geq 0$ . Hence, the fold avoids  $\{F_1 \geq 0, F_0 \geq 0\}$ . Along the physical branch  $F_1 = K_1$ , we therefore have

$$\frac{d}{dx}F_0(x, y(x)) \neq 0 \quad \text{whenever } F_0(x, y(x)) \geq 0.$$

To finish, define

$$f_a(x) = \Delta(x) + a \ln x.$$

The physical branch is characterised by

$$w(x) = \exp\left(\frac{K_1 - f_a(x)}{C}\right) < 1,$$

or, equivalently,  $f_a(x) > K_1$ . Since

$$f'_a(x) = \frac{x^2 + 2ax - 1}{2x^2},$$

and the numerator is strictly increasing on  $(0, 1)$ ,  $f_a$  decreases and then, if its critical point lies in  $(0, 1)$ , increases. Together with  $f_a(0^+) = +\infty$  and  $f_a(1) = 0$ , this shows that the physical  $x$ -domain is an interval  $(0, x_*)$ .

As  $x \downarrow 0$ ,  $w(x)$  tends to zero faster than every positive power of  $x$ , and  $u(x) = xw(x)$ . Consequently,

$$F_0(x, y(x)) = -axw(x) - bw(x) + O(w(x)^2) < 0$$

for all sufficiently small  $x$ .

Now set  $\varphi(x) = F_0(x, y(x))$ . On any component of  $\{\varphi > 0\}$ , the derivative  $\varphi'$  cannot vanish. Such a component must be entered from below, because  $\varphi < 0$  near the left endpoint and a zero of  $\varphi$  cannot have zero derivative. Thus,  $\varphi$  is strictly increasing on that component and cannot subsequently return to zero. There is at most one such component, and every level  $K_0 \geq 0$  is attained at most once. This proves uniqueness.  $\square$

**6.1. Applications.** By the swap symmetry (17), we may assume  $\mu_w \geq \mu_u$ . We first use  $(\mu_u, \mu_w)$  as the anchor in Theorem 6.2.

**Theorem 6.4** (Contact-plane lower bound). *Take arbitrary  $0 < \mu_u < \mu_w < 1$ , and let  $d = \mu_w - \mu_u$ . Let  $A, D$  solve*

$$\begin{pmatrix} -\ln \mu_u & -\ln \mu_w \\ -\ln(1 - \mu_u) & -\ln(1 - \mu_w) \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} d^2/(2\mu_u\mu_w) \\ d^2/[2(1 - \mu_u)(1 - \mu_w)] \end{pmatrix}. \quad (54)$$

Suppose that

$$A \geq 0, \quad D \geq 0.$$

Then

$$\zeta(\mu_u, \mu_w, e_u, e_w) \geq L_{3a} := j(\mu_u, \mu_w) + A[h(\mu_u) - e_u] + D[h(\mu_w) - e_w].$$

The condition  $A, D \geq 0$  is equivalent to

$$\frac{-\ln \mu_w}{-\ln(1 - \mu_w)} \leq \frac{(1 - \mu_u)(1 - \mu_w)}{\mu_u\mu_w} \leq \frac{-\ln \mu_u}{-\ln(1 - \mu_u)}. \quad (55)$$

In particular, it holds, with  $A, D > 0$ , whenever

$$0 < \mu_u \leq \frac{1}{2} \leq \mu_w < 1, \quad \mu_u < \mu_w.$$

*Proof of Theorem 6.4.* First, the matrix in (54) is invertible. Indeed, the function

$$t \mapsto \frac{-\ln t}{-\ln(1 - t)}$$

is strictly decreasing on  $(0, 1)$ . For example, differentiating its logarithm gives

$$-\frac{1}{t(-\ln t)} - \frac{1}{(1 - t)(-\ln(1 - t))} < 0.$$

Since  $\mu_u < \mu_w$ , the determinant of the matrix is positive.

Cramer's rule shows that  $A, D \geq 0$  if and only if the ratio of the two right-hand sides lies between the corresponding column ratios. That ratio is

$$\frac{d^2/(2\mu_u\mu_w)}{d^2/[2(1 - \mu_u)(1 - \mu_w)]} = \frac{(1 - \mu_u)(1 - \mu_w)}{\mu_u\mu_w}.$$

This proves the equivalence with (55).



Under this condition, Theorem 6.2 gives

$$j(x, y) \geq c(y - x) - Ah(x) - Dh(y).$$

Integrating against a representing law gives

$$\mathbb{E}j(U, W) \geq c(\mu_w - \mu_u) - Ae_u - De_w.$$

The contact identity

$$c(\mu_w - \mu_u) = j(\mu_u, \mu_w) + Ah(\mu_u) + Dh(\mu_w)$$

therefore yields

$$\mathbb{E}j(U, W) \geq j(\mu_u, \mu_w) + A[h(\mu_u) - e_u] + D[h(\mu_w) - e_w].$$

Taking the infimum proves the cap bound.

It remains to verify the stated cross-half applicability. For  $0 < t < 1/2$ , the function

$$F_0(t) = -t \ln t + (1 - t) \ln(1 - t)$$

is strictly concave, since

$$F_0''(t) = -\frac{1 - 2t}{t(1 - t)} < 0,$$

and it vanishes at 0 and 1/2. Hence,

$$-t \ln t > -(1 - t) \ln(1 - t).$$

Write  $\mu_w = 1 - v$ , where  $0 < v \leq 1/2$ . For  $\mu_u \leq 1/2$ ,

$$\frac{-\ln \mu_u}{-\ln(1 - \mu_u)} \geq \frac{1 - \mu_u}{\mu_u}, \quad \frac{-\ln \mu_w}{-\ln(1 - \mu_w)} = \frac{-\ln(1 - v)}{-\ln v} \leq \frac{v}{1 - v}.$$

Also,

$$\frac{v}{1 - v} \leq \frac{(1 - \mu_u)v}{\mu_u(1 - v)} \leq \frac{1 - \mu_u}{\mu_u}.$$

The middle expression is precisely  $(1 - \mu_u)(1 - \mu_w)/(\mu_u \mu_w)$ . If  $\mu_u < \mu_w$ , the resulting comparisons with the two logarithmic ratios are strict, including the cases in which exactly one contact coordinate is 1/2. Cramer's rule therefore gives  $A, D > 0$ .  $\square$

The next proposition uses a different anchor point. Suppose  $M = (\mu_u, \mu_w, e_u, e_w)$  admits a triple  $(p, u, v)$  with  $p \in (0, 1]$  and  $u, v \in (0, \frac{1}{2})$  such that

$$\mu_w - \mu_u = p(1 - u - v), \quad e_u = p h(u), \quad e_w = p h(v). \quad (56)$$

Using  $(u, 1 - v)$  as the anchor in Theorem 6.2 yields the following lower bound.

**Proposition 6.5** (Tangent lower bound). *If (56) has a solution  $(p, u, v)$  with  $p \in (0, 1]$  and  $u, v \in (0, \frac{1}{2})$ , then  $(p, u, v)$  is unique and*

$$\zeta(M) \geq L_{3b} := p j(u, 1 - v). \quad (57)$$

*Equality holds in (57) whenever  $pu \leq \mu_u \leq pu + 1 - p$ .*

*Proof. Uniqueness of the contact triple.* Any admissible  $p$  must belong to

$$\mathcal{P} = \left( \frac{\max\{e_u, e_w\}}{\ln 2}, 1 \right].$$

For  $p \in \mathcal{P}$ , define

$$u(p) = \iota(e_u/p), \quad v(p) = \iota(e_w/p), \quad H(p) = p(1 - u(p) - v(p)).$$

The entropy equations in (56) force these values of  $u$  and  $v$ . Differentiating gives

$$u'(p) = -\frac{e_u}{p^2 J(u(p))}, \quad v'(p) = -\frac{e_w}{p^2 J(v(p))},$$

and hence

$$H'(p) = 1 - u(p) - v(p) + \frac{e_u}{pJ(u(p))} + \frac{e_w}{pJ(v(p))} > 0.$$

Thus, the remaining equation  $H(p) = \mu_w - \mu_u$  has at most one solution.

*Equality case.* When  $pu \leq \mu_u \leq pu + 1 - p$ , choose  $\alpha \in [0, 1]$  so that the three-atom law

$$(U, W) \in \{(0, 0), (u, 1 - v), (1, 1)\}$$

with respective masses  $q(1 - \alpha)$ ,  $p$ , and  $q\alpha$ , where  $q = 1 - p$ , matches all four moments and attains  $pj(u, 1 - v)$ . This gives  $\zeta(M) = L_{3b}$ .  $\square$

## 7. CONCLUSION

In this short expository note, we give the details of the three lower bounds used in the proof of the validity of the hybrid Bellman candidate in the full version.

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